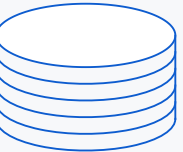


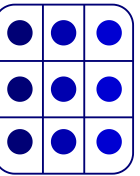
Data Sources & Preprocessing



Five Binary Sentiment Datasets

- SST-2 (EN)
- NSMC (KO)
- CineXDrama (BN)
- HausaMovieReview (HA)
- DravidianCodeMix (ML)

Common Preprocessing



Data Splits

Train / Dev / Test

Fixed Experiment Bank Construction



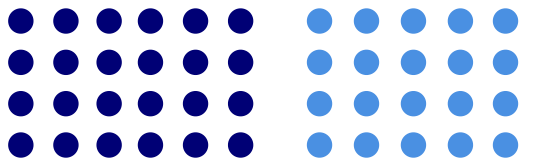
Frozen Training Split Per Dataset

Allocate Base Examples by Class

Negative

Positive

Base Subset (B = 280)



Training-set class ratio

Construct Fixed Experiment Bank

Each class count $\times 5$
→ 1,400 real examples

Synthetic Candidate Pool



For each of the 1,400 bank rows



20 candidates (Gemma 4 31B)



Embeddings (Qwen + BGE)



Seed-synthetic cosine similarity



28,000 requests (1,400 $\times$ 20)

Paired Training Experiments



50 Repetition seeds



11 Augmentation ratios r

A. Base (B only)



same Base within each repetition

B. Matched All-real (B + rB real)



matched total size and class counts

C. Hybrid (B + rB synthetic)



matched total size and class counts

Evaluation Pipeline



Similarity Selection

11 augmentation ratios

$\times$

17 similarity-filter

select similarity-unfiltered



Weighting Search

11 augmentation ratios

$\times$

7 weighting levels



OOF Decision Freeze

Base/Hybrid + threshold
training-only OOF



One-shot Test

frozen policy once
no test-set retuning